

SEMESTER S4

PATTERN RECOGNITION

(Common to CS/CM/CA/AM/CN/CI)

Course Code	PECST412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GAMAT101, GAMAT201, GAMAT301, PCCST303	Course Type	Theory

Course Objectives:

1. To introduce a foundational understanding of the fundamental principles, theories, and methods used in pattern recognition.
2. To develop practical skills in implementing pattern recognition algorithms and techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Foundations of Pattern Recognition Introduction to Pattern Recognition - Definitions and applications of pattern recognition, Overview of pattern recognition systems (Text 2, Chapter 1) Statistical Pattern Recognition - Bayes decision theory, Parametric methods: Maximum likelihood estimation, Bayesian estimation (Text 1, Chapters 1, 2) Non-Parametric Methods - k-Nearest neighbors, Parzen windows (Text 2, Chapter 4)	9
2	Feature Extraction and Selection Feature Extraction - Importance of feature extraction, Techniques for feature extraction: PCA, LDA, Feature extraction in image and signal processing (Text 1, Chapter 3) Feature Selection - Importance of feature selection, Techniques for feature	9

	selection: filter methods, wrapper methods, Feature selection criteria (Text 2, Chapter 6)	
3	Supervised and Unsupervised Learning Supervised Learning - Basics of supervised learning, Linear classifiers: perceptron, logistic regression, Support vector machines (SVM) (Text 1, Chapter 4) Unsupervised Learning - Basics of unsupervised learning, Clustering techniques: k-means, hierarchical clustering, Gaussian Mixture Models (GMM) (Text 1, Chapter 9)	9
4	Advanced Topics and Applications Hidden Markov Models (HMMs) - Basics of HMMs, HMM for sequence modeling, Applications of HMMs in speech and language processing (Text 1, Chapter 13) Ensemble Methods - Basics of ensemble methods, Bagging, boosting, and random forests, Applications and case studies (Text 1, Chapter 14) Applications and Case Studies - Real-world applications of pattern recognition, Case studies in image and speech recognition, Future trends in pattern recognition (Text 2, Chapter 10)	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and Explain fundamental Concepts of Pattern Recognition:	K2
CO2	Apply Classification and Clustering Techniques:	K3
CO3	Implement Feature Extraction and Dimensionality Reduction Techniques	K3
CO4	Apply Statistical and Non-Parametric Methods for Pattern Recognition	K3
CO5	Develop Solutions for Real-World Pattern Recognition Problems and Analyze Case Studies:	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3		3							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3
CO5	3	3	3			3		3				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pattern Recognition and Machine Learning	Christopher M. Bishop	SPRINGER	1/e, 2009
2	Pattern Classification	Richard Duda, Peter Hart, David Stork	Wiley	2/e, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Nature of Statistical Learning Theory	Vladimir Vapnik	Springer-Verlag New York Inc.	2/e, 2010
2	The Elements of Statistical Learning	Jerome Friedman, Robert Tibshirani, Trevor Hastie	Springer-Verlag New York Inc	9/e, 2017
3	Pattern Recognition	S.Theodoridis and K.Koutroumbas	Academic Press	4/e, 2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105101/
2	https://archive.nptel.ac.in/courses/117/105/117105101/
3	https://archive.nptel.ac.in/courses/117/105/117105101/
4	https://archive.nptel.ac.in/courses/117/105/117105101/